CLASSIFICATION & DECISIONING / MLOPS

From churn predictions to repeatable retention decisions

Deliver cost-aware retention recommendations through an API, monitor quality and keep model changes controlled.

WHY MOVE BEYOND THE NOTEBOOK?

Reliable delivery

Churn scores and retention recommendations for downstream applications.

Less manual work

Repeatable checks, training and releases with traceable results.

Controlled updates

Monitor quality, evaluate candidates and verify the rollback path.

Release checks

Candidates must pass defined checks before promotion.

1 of 3 promoted

Three retraining events; one model promotion.

Rollback verified

100% traffic restored; readiness HTTP 200.

RETRAINING IS NOT AN AUTOMATIC MODEL REPLACEMENT

New candidates are evaluated before promotion. Only an approved candidate can replace the active model.

01 Train candidate

Create a new model version.

02 Check candidate

Evaluate against the active model.

03 Release & verify

Publish release and verify serving.

If promotion checks fail, the current Champion remains active.

Version	Registered at	Created by	Tags	Aliases
Version 5	10/06/2026, 07:27:48 AM		model_type: gradient_boosting deployed_at: 2026-10-06 05:27:48.68587	@ champion
Version 4	10/06/2026, 07:18:38 AM		model_type: gradient_boosting deployed_at: 2026-10-06 05:18:40.41307	@ challenger

Local MLflow evidence: versioned candidates and explicit Champion / Challenger aliases. The controlled drift experiment recorded three retraining events and one promotion.

TESTED RECOVERY AFTER AN APPLICATION UPDATE

Cloud Run rollback

- Environment: `prod`
- Service: `mlops-churn-prediction-prod-api`
- Revision: `mlops-churn-prediction-prod-api-00001-b2v`
- Traffic: `100%`
- Readiness: `HTTP 200`

FastAPI, Prefect, private MLflow on Cloud Run, Cloud SQL, GCS, Terraform and GitHub Actions.

Independent reference project using public Telco data and documented synthetic target drift. Retention outcomes use assumed intervention effects. Cloud deployment and rollback successfully verified.

Have a working model? Start with a Notebook-to-MLOps Assessment.

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